

Given: Due:	Further Maths (Decision) HW 27	MARK SCHEME														
Chapter 8: Critical Path Analysis																
Q1. (a) Draw the activity network described in this precedence table, using activity on arc and exactly two dummies. (4) (b) Explain why each of the two dummies is necessary. (3)	<table><tr><th>Activity</th><th>Immediately preceding activities</th></tr><tr><td>A</td><td>-</td></tr><tr><td>B</td><td>-</td></tr><tr><td>C</td><td>A</td></tr><tr><td>D</td><td>B</td></tr><tr><td>E</td><td>B, C</td></tr><tr><td>F</td><td>B, C</td></tr></table>	Activity	Immediately preceding activities	A	-	B	-	C	A	D	B	E	B, C	F	B, C	
Activity	Immediately preceding activities															
A	-															
B	-															
C	A															
D	B															
E	B, C															
F	B, C															

Whilst some very good answers were seen, particularly in part (a) this question proved challenging for many. Common errors in part (a) included: using activity on vertex; lack of arrows; not having a single start and a single finish and most common of all an incorrect second dummy. Many candidates were able to correctly draw activities A to D and place the first dummy but many either omitted the second dummy completely or placed it parallel to the first, but opposite in direction. Similarly in part (b) most candidates were able to explain the need for the first dummy, although the words 'precede' and 'proceed' were either confused or misunderstood. The reason for the second dummy was often omitted, of those who did attempt it the reason was often thought due to their being independent rather than needing to have a unique representation in terms of events.

- (a) **1M1** Activity on arc, all activities (A-F) present, condone lack of events
1A1 A,B,C,D and first dummy correct place, one start.
2A1 E, F and second dummy correct place (on E or F)
3A1 All arrows – including on dummies, one finish
 (b) **1B1** Dummy 1 correctly justified – give bod [ref to B, C, D and E/F]
2B1 Dummy 2 correctly justified – give bod
3B1 A bonus for two good answers

Q2. A project is modelled by the activity network shown in Figure 5. The activities are represented by the arcs. The number in brackets on each arc gives the time, in hours, to complete the activity. Some of the early and late times for each event are shown.

(a) Calculate the missing early and late times and hence complete Diagram 1 in your answer book.

(b) Calculate the total float on activities D, G and I. You **must** make your calculations clear.

(c) List the critical activities.

Each activity requires one worker.

(d) Calculate a lower bound for the number of workers needed to complete the project in the minimum time.

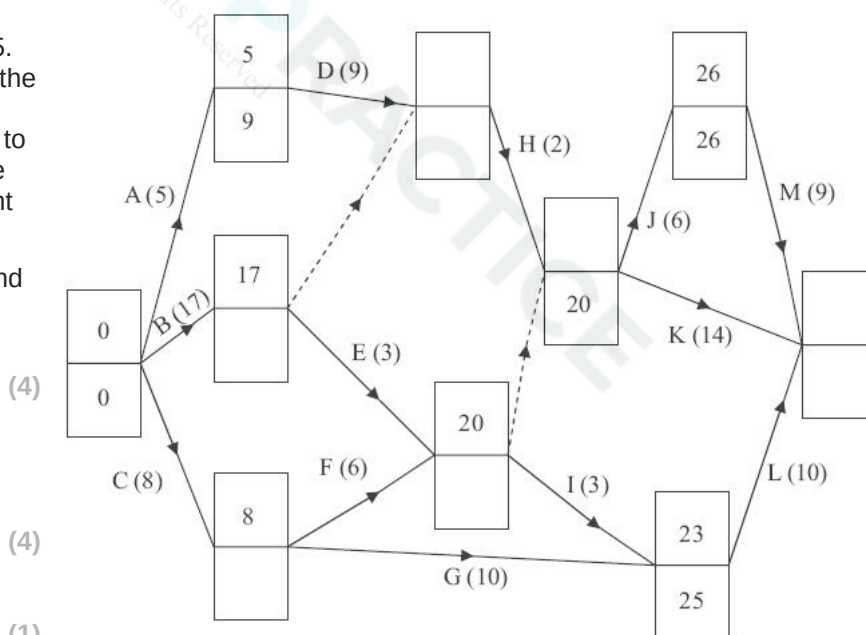


Figure 5

This question was usually well done, and a rich source of marks for most candidates. Most were able to gain all four marks in part (a), but the dummies caused problems for some, leading to incorrect numbers at the start of activities E and H. Many calculated the floats in part (b) correctly and showed the three numbers they used to calculate each float as requested, others wasted time creating a table of all the early and late start and finish times and the durations, for all activities, but then did not indicate which numbers they had used to find the floats, so losing marks. Part (c) was usually well-answered but occasionally K was added to the list of critical activities. Most candidates were able find a lower bound of 3, but some attempted a scheduling or tried to draw a Gantt chart, again wasting time, others used an incorrect method such as 35/13.

Question Number	Scheme	Marks
(a)		m1 A1 m1 A1 (4)
(b)	Total float on D = $18\sqrt{5} - 9 = 4\sqrt{5}$ G = $25 - 8 - 10 = 7$ I = $25 - 20 - 3 = 2$	m1 A2, 1, 0 B1 (4)
(c)	Critical activities: B E J m	B1 (1)
(d)	Lower bound = $\frac{102}{35} = 2.914$ $\therefore 3$ workers	m1 B1 (2)

- (a) 1M1 Top 3 boxes completed, generally ascending L to R
1A1 cao
2M1 Bottom 5 boxes completed, generally descending R to L
2A1 cao
- (b) 1M1 Correct (ft) three numbers visible for at least one calculation.
1A1ft one correct value (ft on D)
1A1 2 correct values
1B1 3 correct values (even if no working seen)
- (c) 1B1 cao (condone 'dummy')
- (d) 1M1= B1 3 workers
1A1= B1 correct method used [$102 \div 35$ ft or 2.9...]

Q3. A project is modelled by the activity network shown in Figure 5. The activities are represented by the arcs. The number in brackets on each arc gives the time, in days, to complete the activity. Each activity requires one worker. The project is to be completed in the shortest possible time.

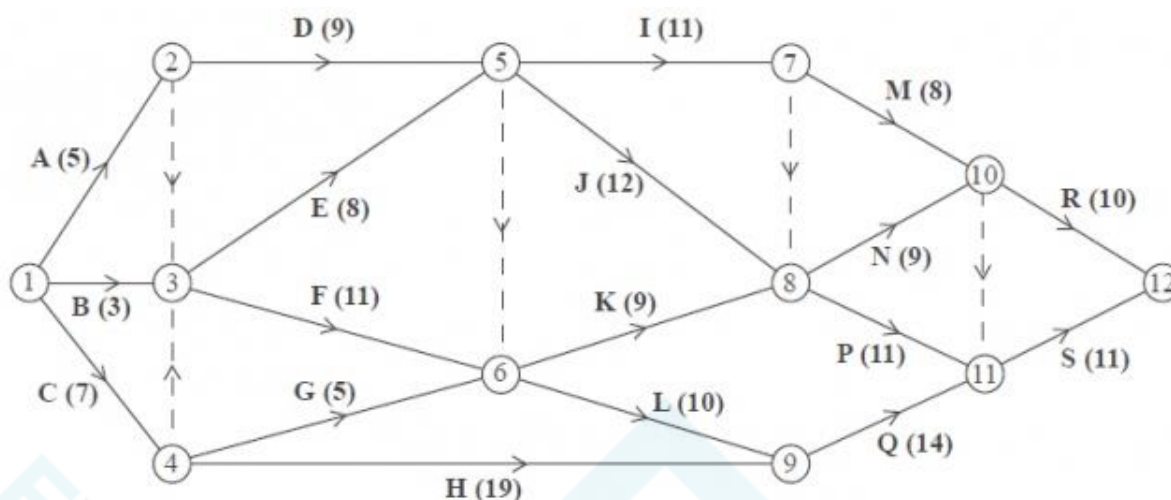


Figure 5

[The sum of the duration of all activities is 172 days]

- (a) Complete Diagram 1 in the answer book to show the early event times and late event times. (4)
 - (b) Calculate the total float for activity M. You must make the numbers you use in your calculation clear. (2)
 - (c) For each of the situations below, explain the effect that the delay would have on the project completion date. (2)
 - (i) A 2 day delay on the early start of activity P.
 - (ii) A 2 day delay on the early start of activity Q.
 - (d) Calculate a lower bound for the number of workers needed to complete the project in the shortest possible time. (1)
- Diagram 2 in the answer book shows a partly completed cascade chart for this project.
- (e) Complete the cascade chart. (4)
 - (f) Use your cascade chart to determine a second lower bound on the number of workers needed to complete the project in the shortest possible time. You must make specific reference to times and activities. (2)
 - (g) State which of the two lower bounds found in (d) and (f) is better. Give a reason for your answer. (2)

This proved to be a challenging and lengthy question for the candidates with only 2.6% scoring full marks. The modal mark was 11 and 23.5% of candidates scored 6 marks or fewer. In part (a), usually the forward pass was carried out correctly. The backward pass was far less successful however, with candidates failing to carry out the pass correctly through the dummies. A high proportion of candidates wrote 9 as the late time at the end of C, 43 at the end of M, 9 or 13 at the end of A and/or 19 at the end of D. In part (b), the majority of candidates knew the method for the float calculation and showed it clearly. Unfortunately, due to an error in part (a), many lost the second accuracy mark.

The delay on activities P & Q were well understood and a lot of correct answers were given in part (c). Some

candidates' responses were more succinct than others and a significant number of candidates described the effect these delays would have on the activity which followed or on the float time and did not as requested, comment on the effect on the project completion date.

In part (d), nearly all candidates gave the correct answer of 4 for the lower bound. A common incorrect answer was 10 arising from dividing by the total number of activities. The cascade chart was relatively challenging to complete in part (e). Adding 11 activities with floats was clearly daunting for weaker candidates who were perhaps running short of time by this stage of the paper. Furthermore the 'odd' times were not marked by vertical lines which added an extra challenge for some. However, most candidates made a very good attempt at this part. Follow through errors from (a) cost accuracy marks and a few omitted one (or more) activities, others who had part (a) correct made minor slips when drawing their activities. A small number ignored the given activities and duplicated them, getting them wrong in some cases. Despite the examples given for A and B there was a variety of different ways of representing activities and floats seen by examiners. A few students had multiple entries on each line and a small number of scheduling diagrams were still seen.

In part (f), there were few completely correct responses. Most candidates opted for an argument involving activities H, I, J, K and L and to score any marks in this part a time had to be provided and for both marks this had to be correct. Several responses failed to score any marks because no time was mentioned at all. Very few candidates gave reasons associated with D, E, F, G and H and those that attempted to do so rarely scored any marks as this was the more challenging of the two options. Many candidates gave arguments based on scheduling which scored no marks.

In part (g) there were very few completely correct responses to this question and it was clear that most candidates were not familiar with the significance of different lower bounds. A significant proportion chose the smaller lower bound "because it is cheaper" or "uses less workers". Of those who selected the higher bound, very few gave the correct reason for doing so. Many candidates misunderstood the question and gave reasons for why the cascade bound was more *reliable*, arguing for example: "because it takes account of when activities must be taking place" or "because it takes account of precedences". Many more, however, gave incorrect and irrelevant reasoning such as "because it is more practical".

Question Number	Scheme	Marks
(a)		M1 A1
		M1 A1 (4)
(b)	Float on M = $42 - 26 - 8 = 8$	M1 A1 (2)
(c)(i)	2 day delay on P – no effect on the project completion date (float on P is 4)	B1
(c)(ii)	2 day delay on Q – project finishes 2 days late (Q is a critical activity)	B1 (2)
(d)	$(172/53 = 3.245, \text{ so})$ a minimum of 4 workers needed	B1 (1)



Question Number	Scheme	Marks
(e)		M1 A1 (any 6 more) M1 A1 (all 11) (4)
(f)	E.g. Activities H, I, J, K and L together with $22 < \text{time} < 26$ stated. So 5 workers needed	M1 A1 (2)
(g)	The cascade gives a higher lower bound, so (f) is better.	M1 A1 (2)
		17 marks

Notes for Question:

Notes:

a1M1: All top boxes complete, values generally increasing left to right, condone one 'rogue' (if values do not increase from left to right then if one value is ignored and then the values do increase from left to right then this is considered to be only one rogue value)

a1A1: CAO.

a2M1: All bottom boxes complete, values generally decreasing right to left, condone one 'rogue'.

a2A1: CAO

b1M1: Correct calculation seen – all three numbers correct (ft), float 0.

b1A1: Float correct (no ft on this mark)

c1B1: CAO

c2B1: CAO

d1B1: 4 with (or without) working scores this mark

e1M1: At least six activities added including six floats. Scheduling diagram scores M0.

e1A1: Six activities including their floats dealt with correctly.

e2M1: All remaining eleven activities including all eleven floats.

e2A1: CAO.

Examples for part (f):

Example 1: Activities H, I, J, K and L with $22 < \text{time} < 26$ so 5 workers needed.

Example 2: At $10 < \text{time} < 14$, F, D, E and H must be happening. Activity G must be happening $7 < \text{time} < 18$ but its duration is 5 so it must also occur at some point in the interval $10 < \text{time} < 14$ so 5 workers needed.

f1M1: Example 1: A statement with the correct number of workers (5) **and** the correct activities (H, I, J, K and L) with some mention of time, **or**

Example 2: A statement with the correct number of workers (5), the correct activities (F, D, E and H) with some mention of time **and** an indication that G **must** be happening with the other four activities at some point - give bod but e.g. 'at time 11 F, D, E, G and H must be happening' is M0. Scheduling the activities only scores M0.

f1A1: A correct, complete full statement with details of both time **and** activities. Candidates only need to give a time within the intervals stated.

Please note strict inequalities for the time. Allow e.g. on 'day 23' as equivalent to $22 < \text{time} < 23$.

g1M1: **Must have attempted both parts (d) and (f).** Their higher lower bound chosen + attempt at a reason.

Allow for the M mark a reason which argues that e.g. the cascade chart gives a better lower bound (e.g. it takes into account exactly when activities must be taking place) **or** e.g. the calculation gives a better lower bound (e.g. as it takes into account the sum of all the activities) but without specifically answering the question of which of the two bounds is better. Give bod on an attempt at a reason.

g1A1: CAO plus a correct reason given. Acceptable reasons e.g. the cascade gives a larger value **or** the bound for the cascade shows that the project cannot be done with fewer workers, etc.